Discrete minus sign uncertainty at odd squares

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Abstract

We prove the odd-integer instances of Conjecture 3.7 under the odd-length formulation of Problem 3.1 in Gonçalves–Oliveira e Silva–Ramos, *New Sign Uncertainty Principles*. Two explicit nonnegative dual certificates give the lower bound; the published sine construction supplies the matching upper bound. The lower bound answers the question posed after Proposition 3.6. The cutoff $\ell - 1$ enters only at the reflected pair $n = \pm(1 - \ell)$.

For odd $\ell \geq 3$, put $N = \ell^2$ and $q = (N-1)/2$. Represent $\mathbb{Z}_N$ by $-q, \dots, q$. Use the unitary transform and full inner product

$$(\mathcal{D}f)(k) = \frac{1}{\ell} \sum_{n \in \mathbb{Z}_N} f(n) e^{-2\pi i n k/N}, \qquad \langle x, y \rangle = \sum_{n \in \mathbb{Z}_N} x(n)\overline{y(n)}.$$

In Problem 3.1 of [1], let $\mathcal{A}_-^{\mathrm{disc}}(q)$ be the real even vectors f satisfying

$$f(0) \geq 0, \qquad (\mathcal{D}f)(0) \leq 0, \qquad f(\pm q), -(\mathcal{D}f)(\pm q) \geq 1.$$

These are the paper's conditions $sf(0), \widehat{f}(0) \leq 0$ and $f(\pm q), s\widehat{f}(\pm q) \geq 1$ with $s = -1$ and $\widehat{f} = \mathcal{D}f$. Evenness makes the conditions at $+q$ and $-q$ redundant. Write k_f for the paper's k_{-f}, namely the smallest nonnegative integer K such that $f(n), -(\mathcal{D}f)(n) \geq 0$ whenever $K \leq |n| \leq q$. Thus, as in Problem 3.1 of [1],

$$\mathbb{A}_-^{\mathrm{disc}}(q) = \min_{f \in \mathcal{A}_-^{\mathrm{disc}}(q)} k_f.$$

For fixed q, tail conditions on $K \leq |n| \leq q$ imply those on $K' \leq |n| \leq q$ for every integer $K' \geq K$. Thus this minimum is also the least feasible cutoff. Before Conjecture 3.7, the authors report numerical confirmation through $\ell = 100$ and ask for a proof of the lower bound.

Theorem 1. *For every odd integer $\ell \geq 3$,*

$$\mathbb{A}_-^{\mathrm{disc}}\big((\ell^2 - 1)/2\big) = \ell.$$

Proof. Suppose some admissible f has $k_f \leq \ell - 1$. By the fixed-q observation, its tail conditions hold at $K = \ell - 1$. Since $\mathcal{D}^2 = I$ on even vectors, $u = (f - \mathcal{D}f)/2$ is real and even, with

$$\mathcal{D}u = -u, \qquad u(0) \geq 0, \qquad u(n) \geq 0 \ (|n| \geq \ell - 1), \qquad u(q) = \frac{f(q) - (\mathcal{D}f)(q)}{2} \geq 1.$$

To exclude this cutoff, use two real even nonnegative dual certificates that vanish wherever u's sign is free. The first, G, is a $+1$ eigenvector; the second, H, has Fourier support in the zeros forced by G and the coset sums.

Write $E_b = \{n \in \mathbb{Z}_N : n \equiv b \pmod{\ell}\}$ and $C_b = \mathbf{1}_{E_b}$ for each fixed integer representative b. Put $\delta = 2\pi/N$, $c(n) = \cos(\delta n)$, $s(n) = \sin(\delta n)$, $P = C_1 + C_{-1}$, $R = C_1 - C_{-1}$ and $t = \tan(\delta/2)$. Define

$$G = (2 + 2c)C_0 + (1 - c)P - tsR. \tag{1}$$

The functions c, s are N-periodic, so G is well defined on $\mathbb{Z}_N$. Finite geometric summation and modulation give, with b fixed,

$$(\mathcal{D}C_b)(k) = e^{-i\delta bk}C_0(k), \qquad \mathcal{D}(e^{\pm i\delta n}g(n))(k) = (\mathcal{D}g)(k \mp 1).$$

Consequently $\mathcal{D}C_0 = C_0$, $\mathcal{D}P = 2cC_0$, $\mathcal{D}(cC_0) = P/2$ and $\mathcal{D}R = -2isC_0$. In particular,

$$\mathcal{D}(cP)(k) = c(k-1)C_1(k) + c(k+1)C_{-1}(k) = (\cos\delta\, cP + \sin\delta\, sR)(k),$$
$$\mathcal{D}(sR)(k) = -s(k-1)C_1(k) + s(k+1)C_{-1}(k) = (\sin\delta\, cP - \cos\delta\, sR)(k).$$

Expanding (1) therefore yields

$$\mathcal{D}G = (2 + 2c)C_0 + P - (\cos\delta + t\sin\delta)cP + (t\cos\delta - \sin\delta)sR = G,$$

using the half-angle coefficient identities

$$\cos\delta + t\sin\delta = 1, \qquad \sin\delta - t\cos\delta = t.$$

On E_0, $G(n) = 4\cos^2(\pi n/N) > 0$ because N is odd. For $n \in E_1$ with $0 \le n < N$,

$$G(n) = \frac{2\sin(\pi n/N)\sin(\pi(n-1)/N)}{\cos(\pi/N)}.$$

Since $\cos(\pi/N) > 0$, $G \ge 0$ there and its only zero is $n = 1$. Both s, R are odd, so G is even. Reflecting gives the same statement on E_{-1} with sole zero -1. Elsewhere $G = 0$. In particular G vanishes at every $1 \le |n| \le \ell - 2$. Unitarity gives $\langle u, G\rangle = \langle \mathcal{D}u, \mathcal{D}G\rangle = -\langle u, G\rangle$, so $\langle u, G\rangle = 0$. Every term is nonnegative: $G(n) > 0$ forces $n = 0$ or $|n| \ge \ell - 1$, where $u(n) \ge 0$. Indeed, nonzero points of E_0 have $|n| \ge \ell$, and the points of $E_{\pm 1} \setminus \{\pm 1\}$ closest to zero are $\pm(1 - \ell)$. Only this reflected pair uses the cutoff $\ell - 1$. Thus u vanishes on E_0 and on $E_{\pm 1}$ except at ± 1.

For $r = 0, \dots, \ell - 1$, let $S_r = \sum_{m=0}^{\ell-1} u(r + m\ell)$. For every $j = 0, \dots, \ell - 1$, with $j\ell$ taken modulo N,

$$0 = -u(j\ell) = (\mathcal{D}u)(j\ell) = \frac{1}{\ell}\sum_{r=0}^{\ell-1} S_r e^{-2\pi i jr/\ell},$$

because $u = 0$ on E_0. Inverting this length-ℓ DFT gives

$$S_r = \sum_{j=0}^{\ell-1} (\mathcal{D}u)(j\ell)e^{2\pi i jr/\ell} = 0.$$

Only $u(1)$ remains in E_1, so $S_1 = u(1) = 0$. Evenness of u then gives $u(-1) = 0$. Hence

$$u = 0 \quad \text{on } E_0 \cup E_1 \cup E_{-1}. \tag{2}$$

The pointwise zeros use both the certificate and the coset sums. For $\ell = 3$ these three classes cover $\mathbb{Z}_9$, already contradicting $u(q) \ge 1$; the following argument applies as well.

Using representatives $0, \dots, N-1$, for $1 \le r \le \ell - 1$ and $0 \le m \le \ell - 1$, define

$$H(r + m\ell) = \cos(\pi/\ell) - \cos\big(2\pi(m + 1/2)/\ell\big) = 2\sin(\pi m/\ell)\sin\big(\pi(m+1)/\ell\big), \tag{3}$$

and set $H = 0$ on E_0. Negation sends (r, m) to $(\ell - r, \ell - 1 - m)$ and preserves the cosine, proving evenness. The sine product proves $H \ge 0$, with zeros precisely at $m = 0, \ell - 1$ in each nonzero class. The row $m = \ell - 1$ has centered coordinate $r - \ell$; the row $m = 0$ has coordinate r. Thus $H = 0$ whenever $|n| \le \ell - 1$. With $a = \cos(\pi/\ell)$ and class-dependent $\beta_r = e^{i\pi/\ell - i\delta r}$, its explicit expansion is

$$H(n) = \sum_{r=1}^{\ell-1} C_r(n) \left(a - \frac{\beta_r}{2} e^{i\delta n} - \frac{\overline{\beta_r}}{2} e^{-i\delta n} \right).$$

Indeed $e^{2\pi i(m+1/2)/\ell} = \beta_r e^{i\delta n}$ on E_r. Each β_r is a scalar in the r-th summand. The formula for $\mathcal{D}C_r$ puts its support in E_0; modulation shifts this support to $E_{\pm 1}$. Thus $\operatorname{supp} \mathcal{D}H \subseteq E_0 \cup E_1 \cup E_{-1}$. By (2) and unitarity,

$$\langle u, H \rangle = \langle \mathcal{D}u, \mathcal{D}H \rangle = -\langle u, \mathcal{D}H \rangle = 0.$$

The vector u can be negative only at $1 \le |n| \le \ell - 2$. There $H = 0$, as also at 0 and $\pm(\ell - 1)$, so every summand is nonnegative. Writing $\ell = 2h + 1$ gives $1 \le h \le \ell - 2$ and $q = h\ell + h$. Thus $r = m = h$ in (3), and $H(q) = \cos(\pi/\ell) - \cos\pi = 1 + \cos(\pi/\ell) > 0$. Hence

$$0 = \langle u, H \rangle \ge u(q) H(q) > 0,$$

a contradiction. The integral cutoff therefore satisfies $\mathbb{A}_-^{\mathrm{disc}}(q) \ge \ell$.

For completeness, the upper bound is the sine construction of Proposition 3.6 in [1]. Its vector $g - \widehat{g}$ is unnormalized; the printed membership statement uses $\mathbb{A}_-^{\mathrm{disc}}(1)$. We rescale below to meet both endpoint inequalities. Put $w(n) = \sin(2\pi|n|/\ell)$ for $|n| < \ell$, and zero elsewhere. This equals the paper's g, since its values at $\pm\ell$ vanish. The term $j = \ell$ in the paper's finite sum likewise vanishes. For $n \ne \pm\ell$ in $[-q, q]$, finite summation gives

$$(\mathcal{D}w)(n) = \frac{\sin(2\pi/\ell)\,[1 - \cos(2\pi n/\ell)]}{\ell[\cos(2\pi n/N) - \cos(2\pi/\ell)]}.$$

At $n = \pm\ell$ the quotient has a double zero in the numerator and a simple zero in the denominator. Its removable values agree with the direct sums

$$(\mathcal{D}w)(\pm\ell) = \frac{1}{\ell} \sum_{j=1}^{\ell-1} \sin(4\pi j/\ell) = 0, \qquad (\mathcal{D}w)(0) = \frac{2}{\ell} \sum_{j=1}^{\ell-1} \sin(2\pi j/\ell) = 0.$$

Also $w(0) = 0$. For $\ell < |n| \le q$, the denominator is negative and the numerator nonnegative. Since $q \equiv h \not\equiv 0 \pmod{\ell}$, $(\mathcal{D}w)(q) < 0$. The explicit vector

$$f_* = \frac{w - \mathcal{D}w}{-(\mathcal{D}w)(q)}$$

satisfies $\mathcal{D}f_* = -f_*$, $f_*(0) = (\mathcal{D}f_*)(0) = 0$, and $f_*(q) = -(\mathcal{D}f_*)(q) = 1$, with a nonnegative ℓ-tail. It is therefore admissible. At $n = \ell - 1$, $w(n) = -\sin(2\pi/\ell) < 0$, whereas $(\mathcal{D}w)(n) > 0$ by the quotient, whose numerator and denominator are positive there. Thus $f_*(\ell - 1) < 0$ and its cutoff is exactly ℓ. This proves $\mathbb{A}_-^{\mathrm{disc}}(q) \le \ell$. □

Scope. The theorem proves all instances of Conjecture 3.7 defined by the odd-length Problem 3.1: integer $q = (\ell^2 - 1)/2$ forces ℓ odd. No even-length extension is treated. It gives $\mathbb{A}_-^{\mathrm{disc}}(q)/\sqrt{2q+1} = 1$ along odd squares. This subsequence agrees with Conjecture 3.4 combined with the known value $\mathbb{A}_-(1) = 1$, the continuous one-dimensional constant discussed in [1]. It does not prove existence of the full limit. The broader feasibility, monotonicity, range and asymptotic assertions belong to Conjectures 3.4–3.5 of [1].

For Poisson summation methods in continuous sign uncertainty, see Cohn–Gonçalves [2, §2.1].

Checker. The standard-library script `check_diogo.py` runs with `python3 -I check_diogo.py`. For $\ell = 3, 5, 7, 9$, it checks the Fourier equalities and supports exactly in a cyclotomic field. Its decimal sign checks are illustrative. The proof above supplies the signs and the result for all odd ℓ. The script's SHA-256 is `5a13c6c907e6e6031980adf53d2f95d8173fd57a6d2f7d66304782fd46f228be`.

References

[1] F. Gonçalves, D. Oliveira e Silva and J. P. G. Ramos, *New Sign Uncertainty Principles*, Discrete Analysis 2023:9, doi:10.19086/da.84266; arXiv:2003.10771v4.

[2] H. Cohn and F. Gonçalves, *An optimal uncertainty principle in twelve dimensions via modular forms*, Invent. Math. 217 (2019), 799–831, arXiv:1712.04438v3.